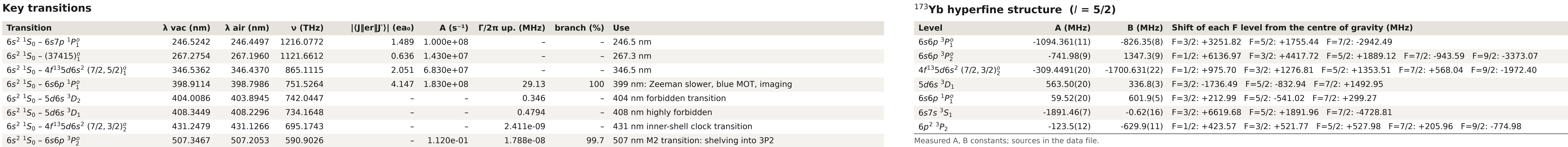
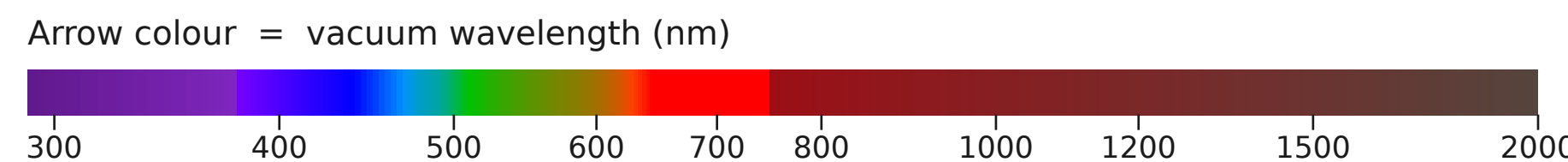


¹⁷³Yb (Yb I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 18 levels, 26 lines.
Yb* ionisation limit $50443.20\text{ cm}^{-1} = 6.25416\text{ eV}$ (198.243 nm)

¹⁷³Yb (Yb I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 18 levels, 26 lines.

Yb⁺ ionisation limit 50443.20 cm⁻¹ = 6.25416 eV (198.243 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f || er || i \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



$6s^2\ ^1S_0 - 6s6p\ ^3P_1^o$	555.8024	555.6480	539.3868	0.540	1.146e+06	0.1824	100	556 nm: green MOT, narrow-line cooling, imaging in tweeze...
$6s^2\ ^1S_0 - 6s6p\ ^3P_0^o$	578.4212	578.2609	518.2944	-	4.800e-02	7.579e-09	100	578 nm optical clock transition

$656p\ ^3P_1 - 6s7s\ ^1S_0$	611.2976	611.1285	490.4198	0.220	4.293e+05	3.475	1.97	611 nm
$656p\ ^3P_0 - 6s7s\ ^3S_1$	649.0871	648.9078	461.8678	1.951	9.400e+06	11.53	13	649 nm clock-state repumper / 3P0 detection
$656p\ ^3P_1 - 6s7s\ ^3S_1$	680.1479	679.9602	440.7754	3.480	2.600e+07	11.53	36	680 nm decay channel of 3S1 during repumping
$656p\ ^3P_0 - 6s7s\ ^3S_1$	770.1606	769.9487	389.2596	5.003	3.700e+07	11.53	50.8	770 nm 3P2 repumper
$656p\ ^3P_1 - 5d6s\ ^3D_2$	1032.4544	1032.1715	290.3687	0.540*	1.074e+05	0.02375	71.9	1032 nm
$656p\ ^1P_1 - 6s7s\ ^1S_0$	1077.3043	1077.0093	278.2802	4.310*	3.010e+07	3.475	100	1077 nm main decay channel of 6s7s 1S0
$656p\ ^3P_2 - 5d6s\ ^3D_2$	1255.1340	1254.7907	238.8530	0.380*	2.959e+04	0.02375	19.8	1255 nm
$656p\ ^1P_1 - 6s7s\ ^3S_1$	1311.2226	1310.8641	228.6358	0.730*	1.596e+05	11.53	0.21	1311 nm weak decay channel of 3S1
$656p\ ^3P_0 - 5d6s\ ^3D_1$	1386.7610	1388.3814	215.8705	2.759	1.920e+06	0.4794	63.8	1389 nm clock-state repumper
$656p\ ^3P_1 - 5d6s\ ^3D_2$	1479.3029	1478.8986	202.6579	4.030*	2.033e+06	0.346	93.5	1480 nm
$656p\ ^3P_1 - 5d6s\ ^3D_1$	1539.1494	1538.7289	194.7780	2.392	1.060e+06	0.4794	35.2	1539 nm decay channel of 3D1
$656p\ ^3P_2 - 5d6s\ ^3D_2$	1798.3949	1797.9039	166.7000	6.120*	1.864e+06	0.2947	100	1798 nm cycling transition from 3P2
$656p\ ^3P_2 - 5d6s\ ^3D_1$	1983.5130	1982.9717	151.1422	2.390*	2.966e+05	0.346	13.6	1983 nm

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	168-174	1887.64(23)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	170-174	1187.8(72)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	171-174	940.8(9)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	172-174	533.0(3)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	173-174	291.51(5)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	176-174	-509.25(7)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	170-174	2531(5)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	171-174	2019(6)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	172-174	1100(4)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	173-174	609(6)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	176-174	-1059(11)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	170-174	2542(12)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	171-174	2038.9(12)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	172-174	1112(8)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	173-174	619.4(8)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	176-174	-1063(6)
$6s^2\ ^1S_0 - 4f^3s^2d6s^2\ (7/2, 3/2)_2^o$	431.248	171-176	-3406.79(3)
$6s^2\ ^1S_0 - 4f^3s^2d6s^2\ (7/2, 3/2)_2^o$	431.248	173-176	-1797.80(26)
$6s^2\ ^1S_0 - 4f^3s^2d6s^2\ (7/2, 3/2)_2^o$	431.248	174-176	-1115.787(25)
$6s^2\ ^1S_0 - 4f^3s^2d6s^2\ (7/2, 3/2)_2^o$	431.248	172-176	-2296.403(25)
$6s^2\ ^1S_0 - 4f^3s^2d6s^2\ (7/2, 3/2)_2^o$	431.248	170-176	-3926.431(26)
$6s^2\ ^1S_0 - 4f^3s^2d6s^2\ (7/2, 3/2)_2^o$	431.248	168-176	-5680.382(43)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$